

Newman-Penrose Formalism Calculations

Stephani et al., Equation (36.16_2)

Metric

$$ds^2 = \frac{1}{V^2} dx_1 \otimes dx_1 + \frac{1}{V^2} dx_2 \otimes dx_2 + \frac{1}{V^2} dx_3 \otimes dx_3 - \frac{M^2}{V^2} dx_4 \otimes dx_4$$

Coordinates

$$(x^\mu) = (x_1, x_2, x_3, x_4)$$

Metric Functions

$$V(x_1, x_2, x_3, x_4) = V$$

$$M(x_4) = M$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}}{2V} \right) dx_1 + \left(\frac{\sqrt{2}M}{2V} \right) dx_4$$

$$n = \left(-\frac{\sqrt{2}}{2V} \right) dx_1 + \left(\frac{\sqrt{2}M}{2V} \right) dx_4$$

$$m = \left(\frac{\sqrt{2}}{2V} \right) dx_2 + \left(\frac{i\sqrt{2}}{2V} \right) dx_3$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2V} \right) dx_2 + \left(-\frac{i\sqrt{2}}{2V} \right) dx_3$$

Contravariant components

$$l = \left(\frac{1}{2} \sqrt{2}V \right) dx_1 + \left(-\frac{\sqrt{2}V}{2M} \right) dx_4$$

$$n = \left(-\frac{1}{2} \sqrt{2}V \right) dx_1 + \left(-\frac{\sqrt{2}V}{2M} \right) dx_4$$

$$m = \left(\frac{1}{2} \sqrt{2}V \right) dx_2 + \left(\frac{1}{2}i\sqrt{2}V \right) dx_3$$

$$\overline{m} = \left(\frac{1}{2} \sqrt{2}V \right) dx_2 + \left(-\frac{1}{2}i\sqrt{2}V \right) dx_3$$

Directional Derivatives

$$D = \frac{1}{2} \sqrt{2}V \partial_{x_1} - \frac{\sqrt{2}V}{2M} \partial_{x_4}$$

$$\Delta = -\frac{1}{2} \sqrt{2}V \partial_{x_1} - \frac{\sqrt{2}V}{2M} \partial_{x_4}$$

$$\delta = \frac{1}{2} \sqrt{2}V \partial_{x_2} + \frac{1}{2}i\sqrt{2}V \partial_{x_3}$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2}V \partial_{x_2} - \frac{1}{2}i\sqrt{2}V \partial_{x_3}$$

Spin Coefficients

$$\rho = \frac{1}{2} \sqrt{2} \frac{\partial}{\partial x_1} V - \frac{\sqrt{2} \frac{\partial}{\partial x_4} V}{2 M}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{2} \sqrt{2} \frac{\partial}{\partial x_1} V + \frac{\sqrt{2} \frac{\partial}{\partial x_4} V}{2 M}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{1}{4} \sqrt{2} \frac{\partial}{\partial x_2} V - \frac{1}{4} i \sqrt{2} \frac{\partial}{\partial x_3} V$$

$$\beta = -\frac{1}{4} \sqrt{2} \frac{\partial}{\partial x_2} V - \frac{1}{4} i \sqrt{2} \frac{\partial}{\partial x_3} V$$

$$\tau = \frac{1}{2} \sqrt{2} \frac{\partial}{\partial x_2} V + \frac{1}{2} i \sqrt{2} \frac{\partial}{\partial x_3} V$$

$$\pi = -\frac{1}{2} \sqrt{2} \frac{\partial}{\partial x_2} V + \frac{1}{2} i \sqrt{2} \frac{\partial}{\partial x_3} V$$

$$\epsilon = -\frac{1}{4} \sqrt{2} \frac{\partial}{\partial x_1} V + \frac{\sqrt{2} \frac{\partial}{\partial x_4} V}{4 M}$$

$$\gamma = -\frac{1}{4} \sqrt{2} \frac{\partial}{\partial x_1} V - \frac{\sqrt{2} \frac{\partial}{\partial x_4} V}{4 M}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{1}{2} V \frac{\partial^2}{(\partial x_1)^2} V - \frac{V \frac{\partial^2}{\partial x_1 \partial x_4} V}{M} - \frac{V \frac{\partial}{\partial x_4} M \frac{\partial}{\partial x_4} V}{2 M^3} + \frac{V \frac{\partial^2}{(\partial x_4)^2} V}{2 M^2}$$

$$\Phi_{01} = \frac{1}{2} V \frac{\partial^2}{\partial x_1 \partial x_2} V + \frac{1}{2} i V \frac{\partial^2}{\partial x_1 \partial x_3} V - \frac{V \frac{\partial^2}{\partial x_2 \partial x_4} V}{2 M} - \frac{i V \frac{\partial^2}{\partial x_3 \partial x_4} V}{2 M}$$

$$\Phi_{02} = \frac{1}{2} V \frac{\partial^2}{(\partial x_2)^2} V + i V \frac{\partial^2}{\partial x_2 \partial x_3} V - \frac{1}{2} V \frac{\partial^2}{(\partial x_3)^2} V$$

$$\Phi_{10} = \frac{1}{2} V \frac{\partial^2}{\partial x_1 \partial x_2} V - \frac{1}{2} i V \frac{\partial^2}{\partial x_1 \partial x_3} V - \frac{V \frac{\partial^2}{\partial x_2 \partial x_4} V}{2 M} + \frac{i V \frac{\partial^2}{\partial x_3 \partial x_4} V}{2 M}$$

$$\Phi_{11} = -\frac{1}{4} V \frac{\partial^2}{(\partial x_1)^2} V + \frac{1}{4} V \frac{\partial^2}{(\partial x_2)^2} V + \frac{1}{4} V \frac{\partial^2}{(\partial x_3)^2} V - \frac{V \frac{\partial}{\partial x_4} M \frac{\partial}{\partial x_4} V}{4 M^3} + \frac{V \frac{\partial^2}{(\partial x_4)^2} V}{4 M^2}$$

$$\Phi_{12} = -\frac{1}{2} V \frac{\partial^2}{\partial x_1 \partial x_2} V - \frac{1}{2} i V \frac{\partial^2}{\partial x_1 \partial x_3} V - \frac{V \frac{\partial^2}{\partial x_2 \partial x_4} V}{2 M} - \frac{i V \frac{\partial^2}{\partial x_3 \partial x_4} V}{2 M}$$

$$\Phi_{20} = \frac{1}{2} V \frac{\partial^2}{(\partial x_2)^2} V - i V \frac{\partial^2}{\partial x_2 \partial x_3} V - \frac{1}{2} V \frac{\partial^2}{(\partial x_3)^2} V$$

$$\Phi_{21} = -\frac{1}{2} V \frac{\partial^2}{\partial x_1 \partial x_2} V + \frac{1}{2} i V \frac{\partial^2}{\partial x_1 \partial x_3} V - \frac{V \frac{\partial^2}{\partial x_2 \partial x_4} V}{2 M} + \frac{i V \frac{\partial^2}{\partial x_3 \partial x_4} V}{2 M}$$

$$\Phi_{22} = \frac{1}{2} V \frac{\partial^2}{(\partial x_1)^2} V + \frac{V \frac{\partial^2}{\partial x_1 \partial x_4} V}{M} - \frac{V \frac{\partial}{\partial x_4} M \frac{\partial}{\partial x_4} V}{2 M^3} + \frac{V \frac{\partial^2}{(\partial x_4)^2} V}{2 M^2}$$

$$\Lambda = -\frac{1}{2} \frac{\partial}{\partial x_1} V^2 + \frac{1}{4} V \frac{\partial^2}{(\partial x_1)^2} V - \frac{1}{2} \frac{\partial}{\partial x_2} V^2 + \frac{1}{4} V \frac{\partial^2}{(\partial x_2)^2} V - \frac{1}{2} \frac{\partial}{\partial x_3} V^2 + \frac{1}{4} V \frac{\partial^2}{(\partial x_3)^2} V + \frac{V \frac{\partial}{\partial x_4} M \frac{\partial}{\partial x_4} V}{4 M^3} + \frac{\frac{\partial}{\partial x_4} V^2}{2 M^2} - \frac{V \frac{\partial^2}{(\partial x_4)^2} V}{4 M^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"